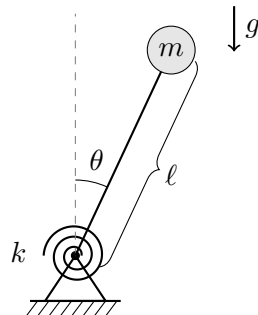


16.07 Dynamics — Quiz 1: Spring-Loaded Inverted Pendulum (Solutions)

1. The nonlinear equation of motion for the system shown is

$$\ddot{\theta} = \frac{g}{\ell} \sin \theta - \frac{k}{m\ell^2} \theta.$$

Find the critical stiffness k_{cr} , in terms of m , g , and ℓ , at which the upright equilibrium transitions from stable to unstable.



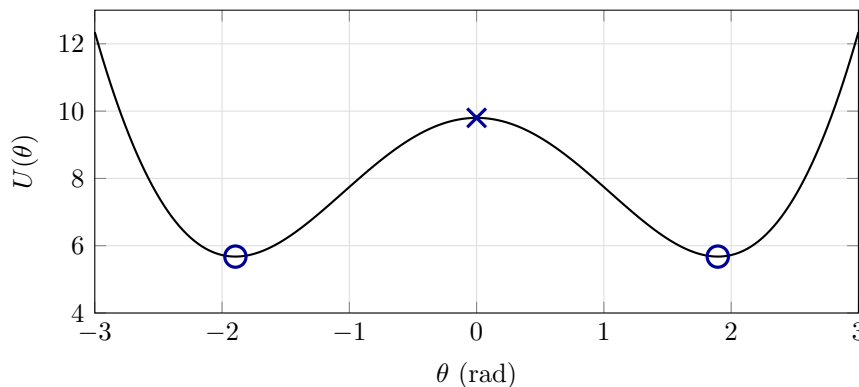
Solution. For small θ , $\sin \theta \approx \theta$, so the linearized equation is

$$\ddot{\theta} + \left(\frac{k}{m\ell^2} - \frac{g}{\ell} \right) \theta = 0.$$

If the coefficient is positive, solutions oscillate at $\omega_n = \sqrt{k/m\ell^2 - g/\ell}$ and the upright equilibrium is stable. If it is negative, solutions grow exponentially and the equilibrium is unstable. The transition happens when the coefficient is zero:

$$\frac{k_{cr}}{m\ell^2} = \frac{g}{\ell} \implies \boxed{k_{cr} = mg\ell}$$

2. The plot below shows $U(\theta)$ for $k = 0.5 k_{cr}$, with $m = \ell = 1$ and $g = 9.8$. Mark every stable equilibrium with a circle (o) and every unstable equilibrium with an $\times$.



Solution. Equilibria are where $dU/d\theta = 0$, corresponding to flat spots or local minima/maxima on the plot above. Local minima, where U is locally convex, are stable, while local maxima, where U is locally concave, are unstable.

3. In general, knowing $\vec{r}$ and $\vec{\tau}$, why can't $\vec{\tau} = \vec{r} \times \vec{F}$ be solved uniquely for $\vec{F}$? Why is it possible to find the spring force this way in this problem?

Solution. The cross product discards any component of $\vec{F}$ parallel to $\vec{r}$: for any scalar λ , $\vec{r} \times (\vec{F} + \lambda \vec{r}) = \vec{r} \times \vec{F}$. Knowing $\vec{r}$ and $\vec{\tau}$ therefore fixes only the component of $\vec{F}$ perpendicular to $\vec{r}$, and there are infinitely many forces that produce the same torque.

In this problem, we know the direction of the equivalent spring force at the mass: a torsion spring acts tangentially, along $\hat{e}_\theta$, which is perpendicular to $\vec{r} = \ell \hat{e}_r$. (Any radial force is carried by the rod and does not enter the θ equation.) With the direction known, only the magnitude remains, and $|\vec{\tau}| = \ell |F|$ gives

$$\vec{F}_{\text{spring}} = -\frac{k\theta}{\ell} \hat{e}_\theta.$$